

Experimental Optimization 2: Optimizing for variable interference

Group 6

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Background

The focus of this project is improving communication reliability in a backscatter system, evaluated using Bit Error Rate (BER) and effective data rate.

In a backscatter system, the tag does not transmit actively. Instead, it reflects and modulates the carrier signal generated by a separate carrier node. The reflected signal is received by a receiver node. The overall path loss depends on both the carrier-to-tag link and the tag-to-receiver link, making the system particularly sensitive to tag position.

The baud rate determines the number of symbols transmitted per second. Each symbol occupies a time window of duration $T_s = 1/f_{baud}$. The receiver must correctly sample the incoming signal within this window to decode each bit. At high baud rates, T_s is short, leaving little margin for correct detection under weak or noisy conditions. Lowering the baud rate increases T_s , giving the receiver more time to correctly detect the incoming signal and reducing the probability of a bit error.

In our system, there is a large amount of interference. This results in a high BER, and lost packages. However, there may sometimes be a 'window of opportunity', where interference is lower than normal. To utilize this, in this optimization we increased the baud rate dramatically, and also increased the packet length. This means that we can send much more data in these small bursts where interference is low, and when interference is higher, the lower baud and safer baseline setup was probably not getting many packets through anyways.

Hypothesis

Increasing the baud rate and packet length, will result in much higher data rates in short bursts. This may be comparable, or even better than, the baseline setup in environments with high interference.

Experimental Setup

The experimental setup consisted of a carrier generator and a receiver node separated by approximately 3 m. The tag was placed at five positions along the line between the carrier generator and receiver.

The tag positions, defined as distance from carrier generator / distance to receiver, were:

- P1: 27 cm / 273 cm
- P2: 50 cm / 250 cm
- P3: 150 cm / 150 cm (midpoint)
- P4: 250 cm / 50 cm
- P5: 277 cm / 23 cm

The midpoint position P3 represents the worst-case scenario for backscatter communication, as both links experience equal and maximum path loss.

Packets were continuously transmitted by the tag. The receiver logged incoming packets until 200 packets were successfully received. The experiment duration Δt was computed as the timestamp difference between the first and last received packet:

$$\Delta t = t_{last} - t_{first}$$

For the midpoint position P3, measurements were continued for up to 5 minutes. If no packets were received within this period, the configuration was recorded as a communication failure with $BER = 100\%$ and $D/R = 0$ byte/s.

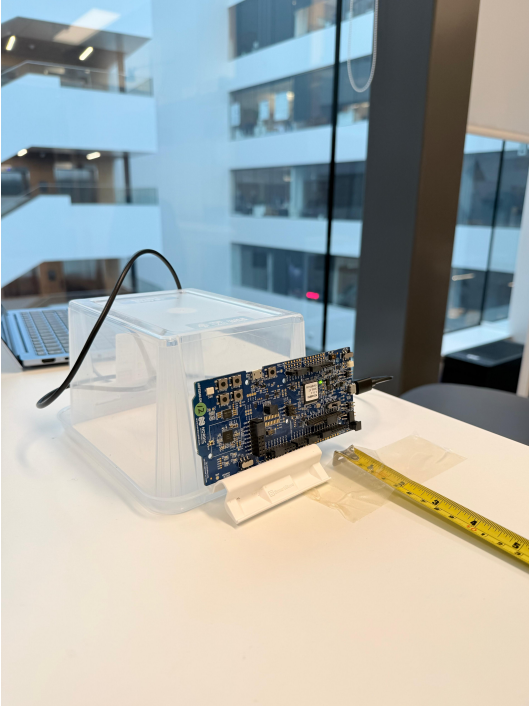


Figure 1: The nRF52840 carrier generator node mounted vertically in a plastic enclosure.

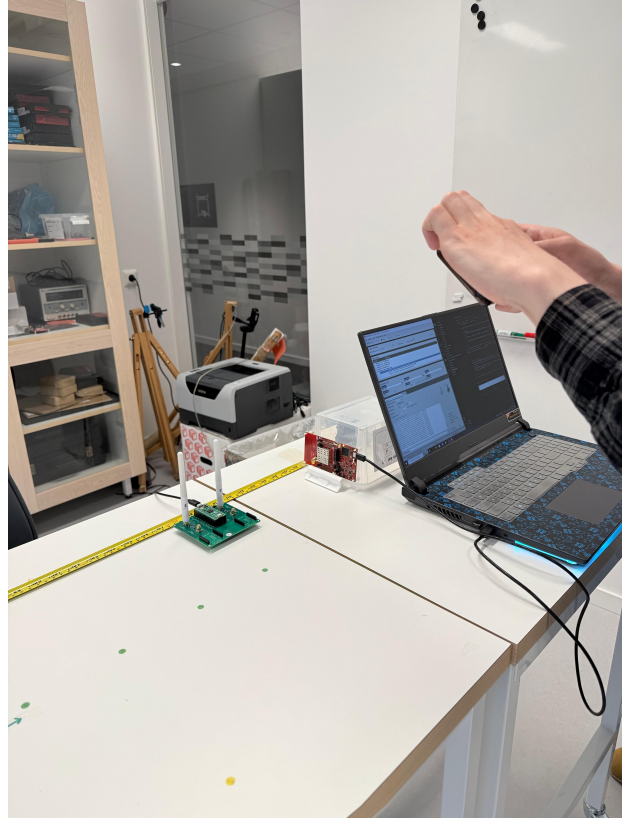


Figure 2: Experimental setup showing the tag (left, green board with antennas) and the CC1352 receiver node (right, red board) connected to a laptop running SmartRF. A measuring tape marks the tag positions.

Metrics

Three metrics were used to evaluate performance.

Bit Error Rate (BER) is the ratio of incorrectly received bits to the total number of received bits:

$$BER = \frac{\text{Number of wrong bits}}{\text{Total number of received bits}}$$

BER is computed only over received packets and does not account for lost packets.

Effective data rate is defined as:

$$D/R = \frac{N_{rx} \times L_{payload}}{\Delta t}$$

where N_{rx} is the total number of packets successfully received and logged by the receiver, $L_{payload} = 12$ bytes is the payload size per packet (excluding the 2-byte pseudo-sequence number field), and Δt is the experiment duration in seconds. The unit is byte/s. Although the given statistics script labels this metric as bit/s, the formula computes $N_{rx} \times 12 \text{ bytes}/\Delta t$, which gives byte/s. This is confirmed by the measured values — for example, baseline P1 gives $205 \times 12/84.42 \approx 29.1$ byte/s, consistent with the recorded value of 29.16. This metric reflects packet loss — missing packets reduce N_{rx} and therefore reduce the effective data rate.

Overall Packet Reception Rate (PRR) Shows the general probability a packet will be received by the receiver, irrespective of its accuracy. This was calculated by first determining the average expected time between packets assuming 100% reception rate, and comparing it with the actual average time between those packets. Expressed differently, this can be seen as the actual time taken for some N packets, over the expected time taken for that same N packets to arrive.

$$PRR = \frac{\text{Actual time taken for N packets}}{\text{Expected time taken for N packets}}$$

Method

The baud rate was set to 250k, and the packet length was increased by 50 percent (from 11 bytes to 16 bytes).

The baseline configuration at 100k baud was repeated three times in a controlled environment and the results were averaged to reduce the effect of environmental variability. All other configurations were measured once. The physical setup and tag positions were kept unchanged across all measurements.

Results

Table 1 shows all individual measurements. For configurations where no packets were received at that location within 15 minutes, BER is recorded as 100% and D/R as 0 byte/s.

Table 1: BER, effective data rate, and experiment duration for each baud rate configuration and tag position. Baseline values are averaged over three trials.

Baud Rate	Position	BER (%)	D/R (byte/s)	Packet Reception Rate (%)
100k (baseline)	P1	10.66	29.16	64.3
	P2	13.03	26.95	58.8
	P3	100.00	0.00	—
	P4	18.51	20.99	45
	P5	15.33	24.55	52.5
250k and increased packet length	P1	33.4	27.87	46.5
	P2	37.7	.93	2
	P3	100.00	0.00	—
	P4	34.5	11.2	1
	P5	100.00	0.00	—

Figure 3 shows the baseline BER per position. Figure 6 shows the BER of the optimization for comparison, at all positions where buckets were recieved. Figure ?? shows the D/R of the optimization for comparison, at all positions where buckets were recieved. Figure ?? shows the PRR of the optimization for comparison, at all positions where buckets were recieved.

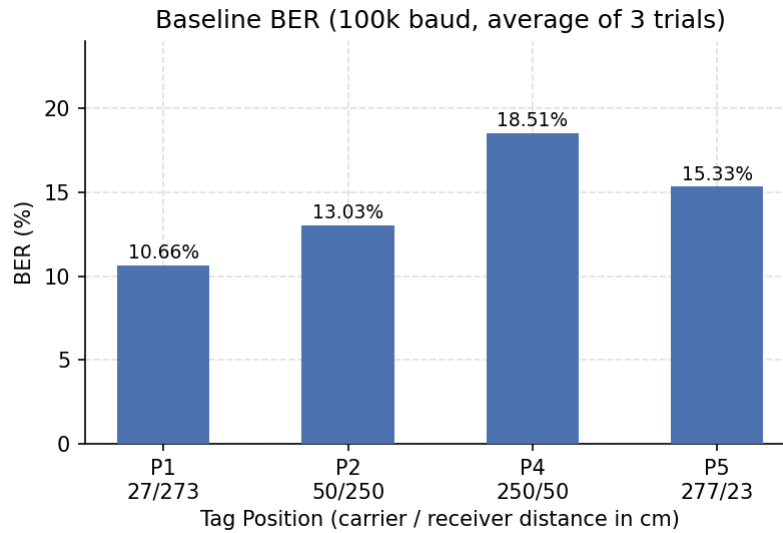


Figure 3: Average BER per tag position for the baseline 100k configuration, averaged over three trials. P3 is excluded as no packets were received within 5 minutes.

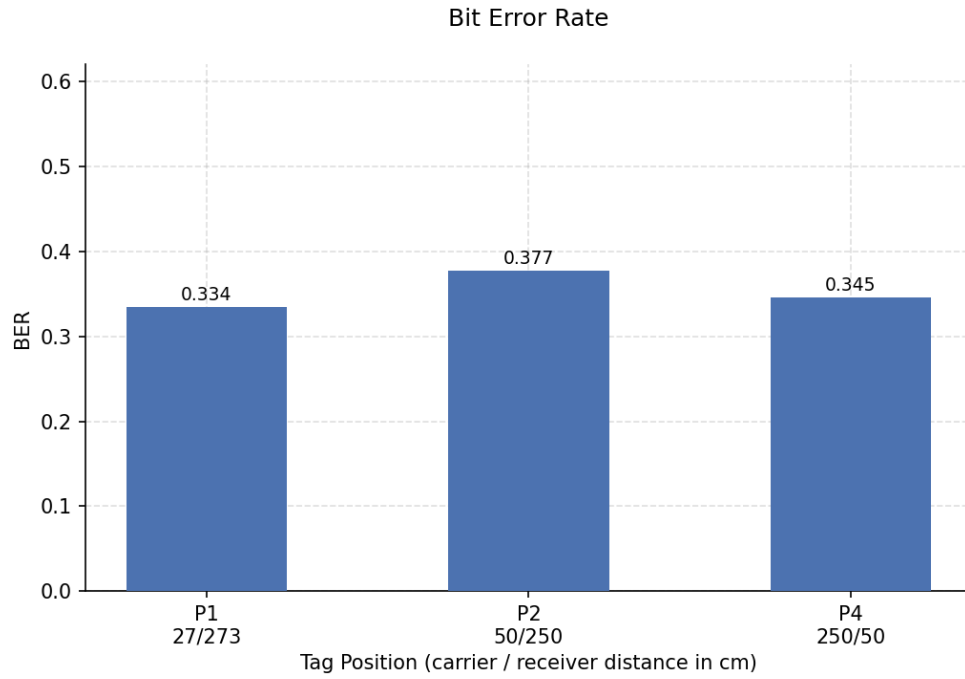


Figure 4: BER per tag position during the optimization experiment. Positions where no packets were received are skipped.

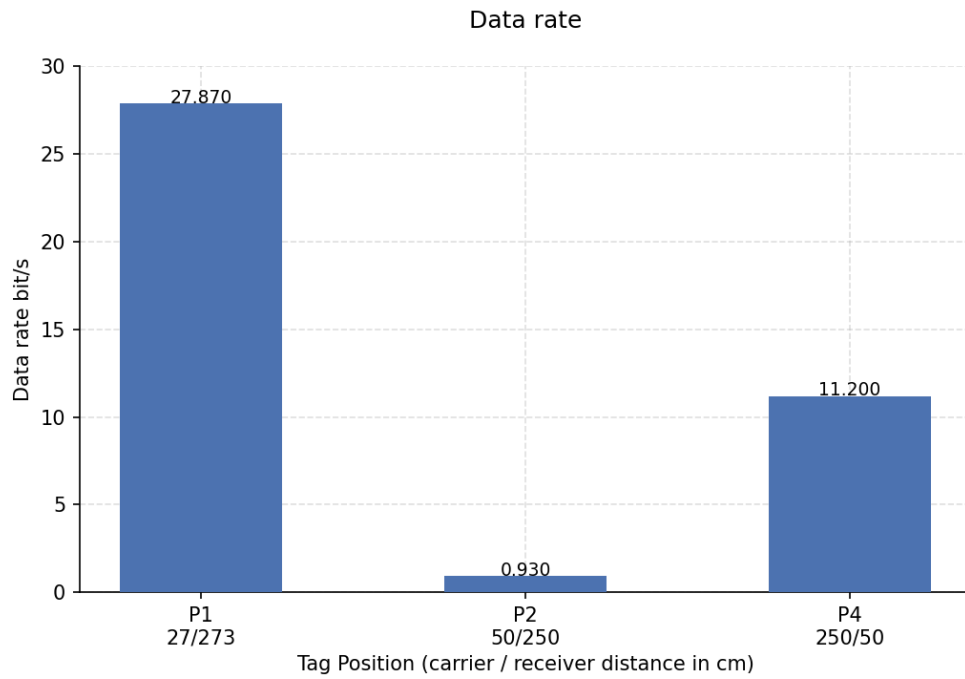


Figure 5: Data Rate per tag position during the optimization experiment. Positions where no packets were received are skipped.

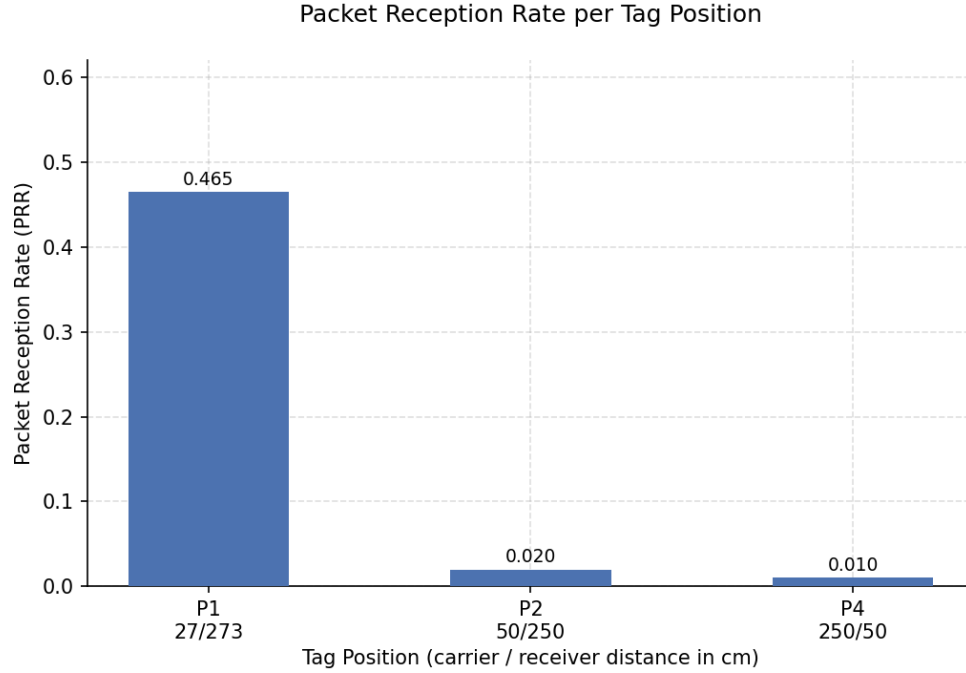


Figure 6: Packet Reception Rate per tag position during the optimization experiment. Positions where no packets were received are skipped.

Discussion and Conclusion

The results are well within expectations. While they did not improve the result relative to the baseline, they did result in the behavior we expected, and may have been better than the baseline in a different environment.

BER The Bit Error Rate did not improve from the baseline to the optimization. In fact, it became significantly worse. However, this optimization is more focused on the data rate, and the nature of this bursty communication during times of low interference.

Data Rate The Data Rate at P1 in the optimization was not far from the data rate in the baseline at a similar position. However, the data rate at any other position was significantly worse than in the baseline. This is likely due to the high drop in energy-per-bit that results from this high of a baud rate, so these further away positions require an even more clear 'window of opportunity', so they do not get to send packets often.

Packet Reception Rate The packet reception rate for any position other than P1 in the optimization was unacceptably low, and in the single digits of percent. This simply implies that the windows of opportunity of low interference are quite rare in our environment, so we could not make the best use of that optimization at those positions.

Conclusion. The results are all well within our expectations. The packet reception came in short, quick bursts rather than slow and more steady streams like in the baseline. While we did not improve performance as intended, this experiment is still interesting, and could potentially be used in a slightly different environment with different interference.